Tomorrow's Leaders

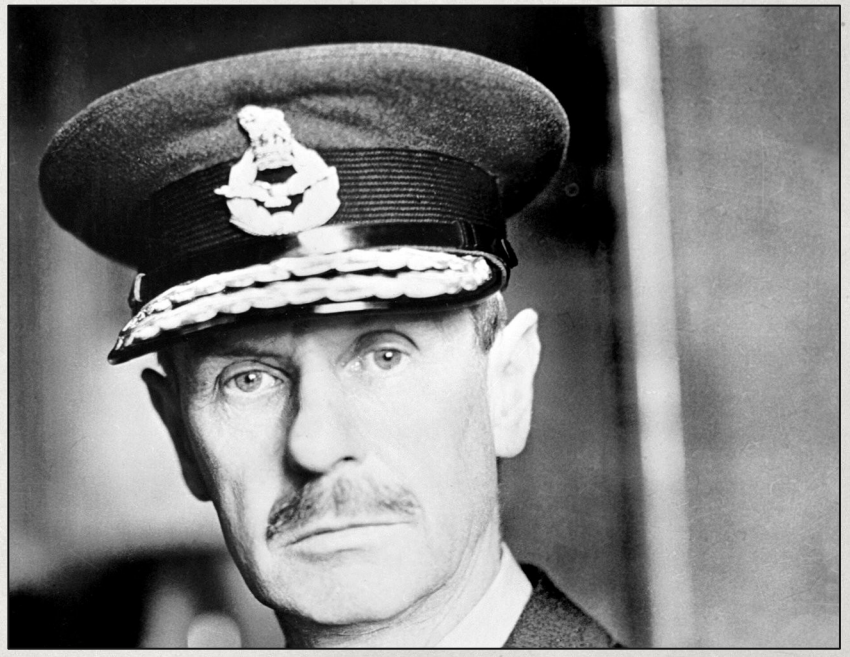
Special edition: effective communication

Hugh Dowding and the Battle of Britain, 1940

# The battle won by information

*How Hugh Dowding's air defence network helped a smaller air force hold off a larger one, and what a student team can learn from it*

By Group [number]: [member names]



*Air Chief Marshal Sir Hugh Dowding, Commander-in-Chief of RAF Fighter Command*
Imperial War Museums, D 1417 (public domain)

# Britain stands alone

*Summer 1940: France has fallen, and the German air force is just across the Channel*



*Spitfires of No. 610 Squadron from Biggin Hill in "vic" formation, 24 July 1940*
Imperial War Museums, CH 740 (public domain)

**10 May** Germany attacks France, Belgium and the Netherlands.

**22 June** France signs an armistice with Germany.

**10 July** The Luftwaffe begins its attack on Britain. Germany cannot invade until it controls the air.

*If you commanded Britain's fighters, where would you send them?*

# Tomorrow's Leaders

*Part 1*

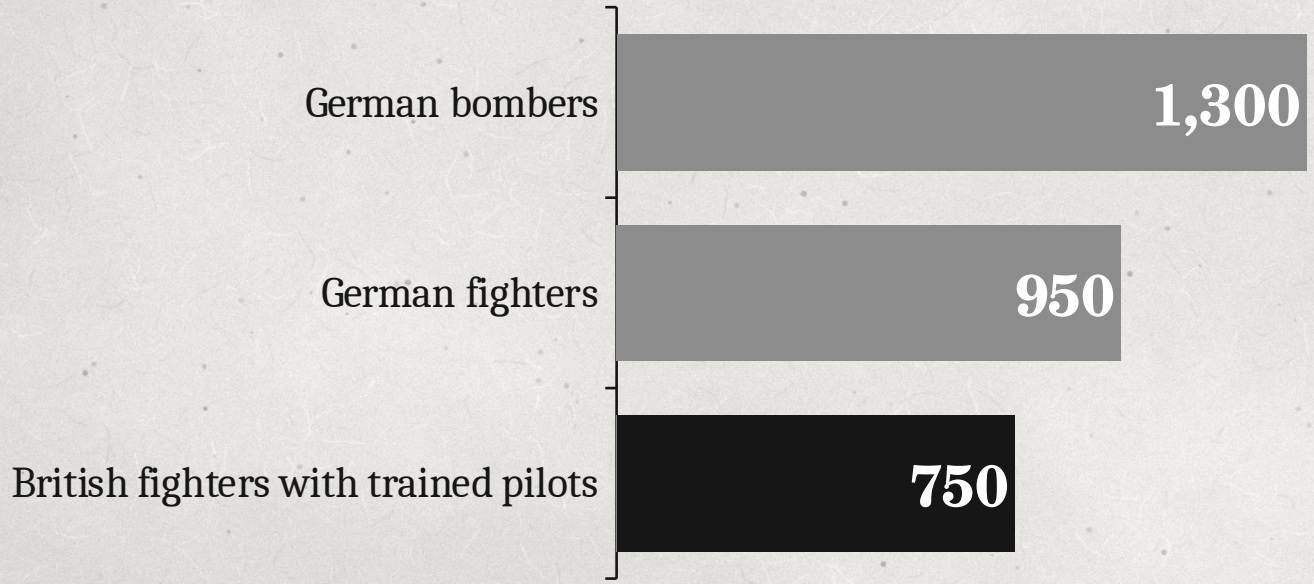
# The story

*How did Britain meet every raid in time, with no fighters to spare?*

1

# 750 British fighters, 2,250 German aircraft

*Strength on 10 August 1940, counting only aircraft ready to fly*




*German numbers are minimums: over 1,300 bombers and over 950 fighters. Britain had 1,100 serviceable fighters, but trained pilots for only about 750.*

*Fighter against fighter, the two sides were close. But Britain's fighters also had to find and stop more than 1,300 bombers.*

*A squadron sent to the wrong place could not reach the real raid in time.*

Gleave (1991, p. 48)

# Twenty minutes to meet each raid

*11 Group warned its controllers that raiders could be over London 20 minutes after the first radar plot*

**20 min**

minutes of warning from a Kent radar station for a raid forming over France

**4 min**

minutes for the network to mark that raid on the operations room table

**2 min**

minutes, at times, for the 11 Group controller to decide

*Every minute spent passing a report along was a minute less for the fighters to reach the raid.*

"The Battle of Britain" (2015); Battle of Britain Bunker (n.d.)

# Seeing the raid was not enough

*A radar operator sees 100 aircraft crossing the Channel. What happens next?*

*Option A*

## Phone the nearest airfield

A few fighters take off alone. Nobody else knows where the raid is going, and other airfields wait for orders.

*Option B*

## Report to headquarters

Each level studies the report and passes an order down. The order reaches the pilots after the bombs have fallen.

*The hard part was not seeing the raid. It was getting the right report to the right person in time.*

Thought experiment based on the timings on slide 5.

# Our group chats have the same problem

*Amy posts in a five-person group chat: "Can someone do the theory part?" What happens next?*

*Option A*

## Nobody replies

Each member thinks someone else will do it. On Thursday, the theory part is still empty.

*Option B*

## Two members both start

They write the same section twice, and nobody starts the comparison table.

*Like the radar operator, Amy has the information. The hard part is getting it to the right person, in a form that person can act on.*

Example from our own group work.

# The man who built the network

*Air Chief Marshal Hugh Dowding, head of RAF Fighter Command from 1936 to 1940*



*Dowding (centre, hands behind his back) as King George VI decorates a pilot at Hornchurch, 27 June 1940*
Imperial War Museums, CH 442 (public domain)

## 1935

As the RAF's head of research, he backs radar after a successful test near Daventry.

## 1936

He becomes the first head of Fighter Command, at Bentley Priory near London.

## 1936 to 1940

He links radar stations, observers, operations rooms and pilots into one network.

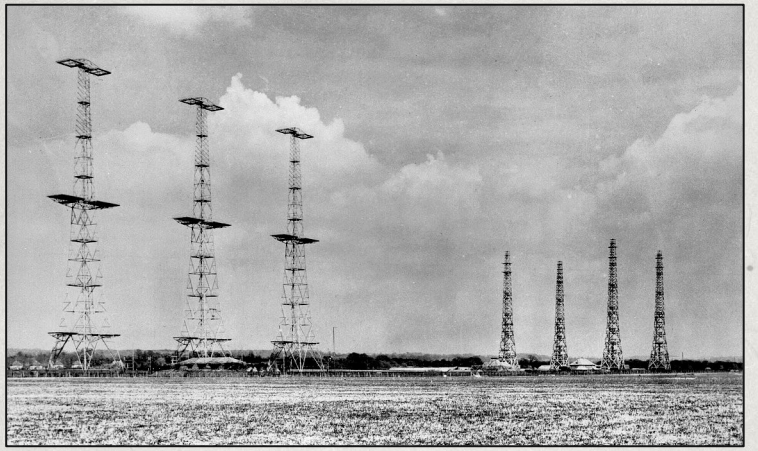
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*His idea, in our words: send each squadron to a raid that someone has already reported, instead of sending it to search the sky.*

Bournemouth University (n.d.); Royal Air Force Museum (n.d.-a); Imperial War Museums (n.d.-b)

# Step 1: Report

*Radar stations covered the sea, volunteers covered the land, and both reported into one network*



*Chain Home radar station at Poling, Sussex, photographed in 1945*
Imperial War Museums, CH 15173 (public domain)



*An observer reports by chest telephone at Kings Langley, 1941 to 1945*
Imperial War Museums, CH 8215 (public domain)

## Chain Home radar

About 20 stations on the coast. They could not track aircraft over land, and their plots were not always accurate.

## Observer Corps

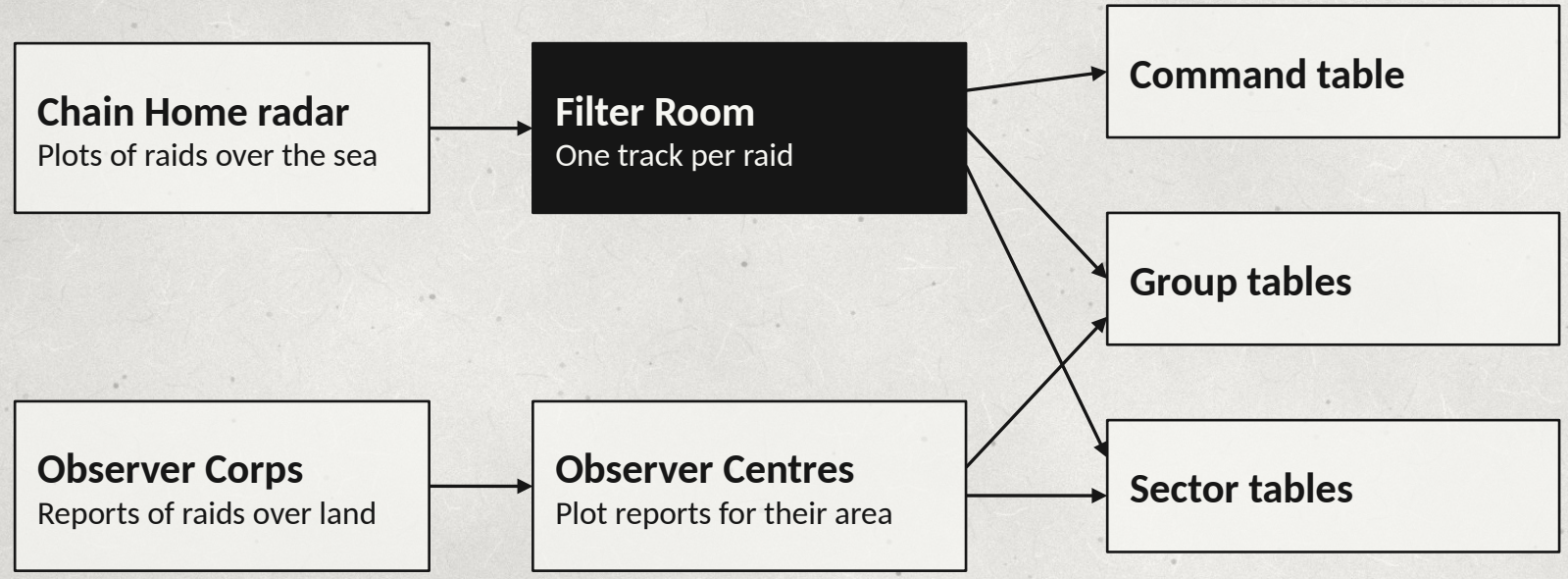
About 30,000 volunteers at about 1,000 posts. They spotted aircraft by eye and reported them by telephone.

***Next problem*** Several radar plots often described the same raid, and some disagreed about its position or strength.

Historic England (2020); Royal Air Force Museum (n.d.-b); Imperial War Museums (n.d.-a)

# Step 2: Filter

*In the Filter Room, staff compared the radar plots and kept one track per raid*




## *At the same time*

Each filtered track went by telephone to the Command, Group and Sector tables at the same moment, so every level saw the same raid.

***Next problem*** Commanders at three levels had to read the same table quickly and know how old each plot was.

Dowding (1946); Imperial War Museums (n.d.-b)

# Step 3: Display

*Every operations table used the same symbols*



*The operations room at Fighter Command headquarters, Bentley Priory (date not recorded)*
Imperial War Museums, C 1869 (public domain)



## Sector clock

Its red, yellow and blue segments each span 5 minutes. Plotters used markers of the same colours, so a controller could see how old each plot was.




## Raid block

A block on the table showed the raid's name and estimated strength, and an arrow showed its direction.




## Tote board

A board showed the state of each squadron, such as “at readiness” or “left ground”.

***Next problem*** A pilot in a cockpit needed his orders in a few words.

Royal Air Force Museum (n.d.-c); Imperial War Museums (2020, n.d.-b)

# Step 4: Order

*A sector controller gave each squadron its course and height by radio, in a few code words*



*Pilots of No. 19 Squadron stage a "scramble" for the camera at Fowlmere, September 1940*
Imperial War Museums, CH 1398 (public domain)

| Code word | Meaning |
|---|---|
| **Scramble** | Take off |
| **Vector 230** | Fly a course of 230 degrees |
| **Angels 20** | Fly at 20,000 feet |
| **Pancake** | Land |

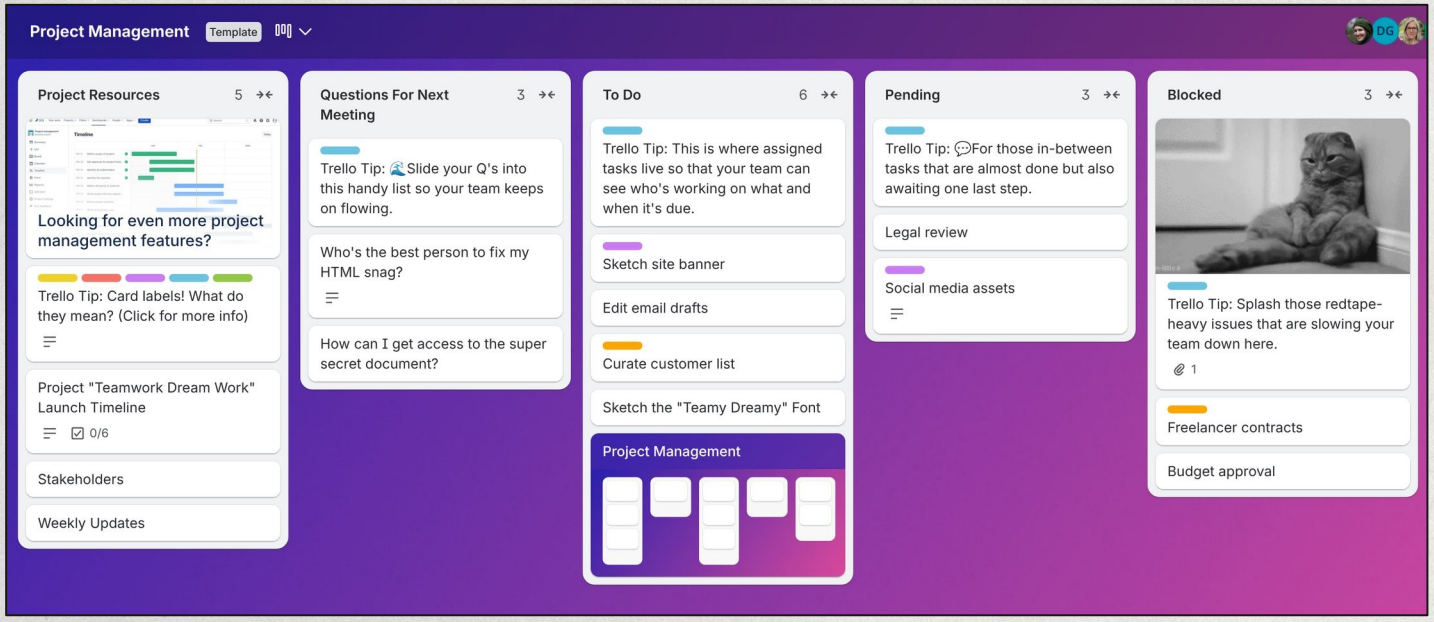
Group decided which sector met each raid and with how many fighters. The sector controller directed the pilots.

***Next question*** Does this table look familiar?

Dowding (1946)

# Look familiar? It is a task board

*Fighter Command's operations table worked like the project boards we use today*




*Trello's Project Management template board, captured 11 November 2026*
Screenshot: trello.com/b/1x4Uql2u

| 1940 | Today |
|---|---|
| One table for every level | **One board for the team** |
| Raid block | **Card** |
| Tote board | **Lists: To Do, Pending, Done** |
| The 20-minute limit | **Due date on the card** |
| Order to one squadron | **Card assigned to one member** |

*Notion, Microsoft Planner and GitHub Projects work the same way.*

Royal Air Force Museum (n.d.-c); Imperial War Museums (2020); Trello (n.d.)

# "We had nothing like it"

*Adolf Galland, a German fighter commander in 1940, on the British network*

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*"From the first the British had an extraordinary advantage, never to be balanced out at any time during the whole war, which was their radar and fighter control network and organisation. It was for us a very bitter surprise. We had nothing like it."*

---

Galland (as cited in Wood, 1991)

# Almost every squadron found its raid

*15 September 1940: two large German daylight attacks on London, and both were beaten off*

**50 of 51** squadrons scrambled that day made contact with the raiders

**56 vs 28** aircraft lost, German against British. 12 RAF pilots were killed.

**17 Sept** Hitler postponed the invasion of Britain until further notice




*The preserved No. 11 Group operations room at Uxbridge, photographed in 2010*
Photo: Danielstirland, CC0, via Wikimedia Commons

*Churchill watched this table that day. "I now asked, 'What other reserves have we?' 'There are none,' said Air Vice Marshal Park."*

Price (2003); Churchill (as cited in Glines, 1990)

# Britain did not have more. It wasted less.

*It wasted less because the right information reached the right person in time*

| | |
|---|---|
| ✓ **Right information** | The Filter Room kept one track per raid and removed duplicate plots. |
| ✓ **Right person** | Each track reached the Command, Group and Sector tables, and Group chose which sector met the raid. |
| ✓ **In time** | A raid appeared on the operations table about 4 minutes after radar detected it. |
| ✓ **In a form they can act on** | "Vector 230, Angels 20" told a pilot where to fly, in a few words. |

**Our definition** Effective communication means the right information reaches the right person in time, in a form they can act on.

**Where it comes from** Keyton (2011): communication transmits "information and common understanding from one person to another". Spitzberg (2013): competence is an impression of effectiveness and appropriateness. Neither mentions time, so we added it from the story.

Keyton (2011, as cited in Lunenburg, 2010); Spitzberg (2013); Dowding (1946); "The Battle of Britain" (2015)

# Tomorrow's Leaders

*Part 2*

## Two theories

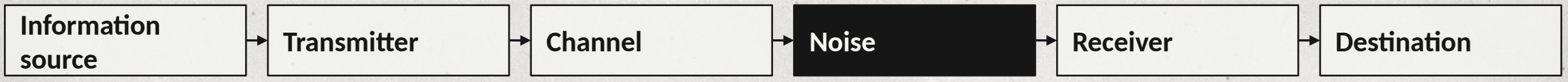
*Why did the network waste so little, and why does Amy's chat waste so much?*

2

# Theory 1: Shannon and Weaver

*A message travels from a source to a destination, and noise can distort it on the way*

*The model*




*In Amy's chat*

| Information source | Transmitter | Channel | Noise | Receiver | Destination |
|---|---|---|---|---|---|
| Amy | Amy types her request | The group chat | Dozens of other messages | Five phones | The member who should act, but nobody knows who |

## Weaver's three levels of communication problems

### A. Technical

Can the symbols be sent accurately?

*Amy's message arrives, but under dozens of others.*

### B. Semantic

Do the symbols carry the intended meaning?

*"The theory part" means different things to different members.*

### C. Effectiveness

Does the meaning change behaviour as intended?

*Nobody starts the work.*

Shannon & Weaver (1949)

# Theory 2: The 7 Cs

*Shannon and Weaver follow the path of a message; the 7 Cs judge the message itself*

## Clarity

What is the one main point?

## Correctness

Are the facts and tone right?

## Conciseness

Can it be said in fewer words?

## Concreteness

Is it backed by specific facts?

## Completeness

Does the reader have all they need to act?

## Consideration

Is it written from the reader's side?

## Courtesy

Is it respectful, even when firm?

*Murphy, Hildebrandt and Thomas (1997) list seven qualities of a good message. We phrase each as a question.*

Murphy et al. (1997)

# Tomorrow's Leaders

*Part 3*

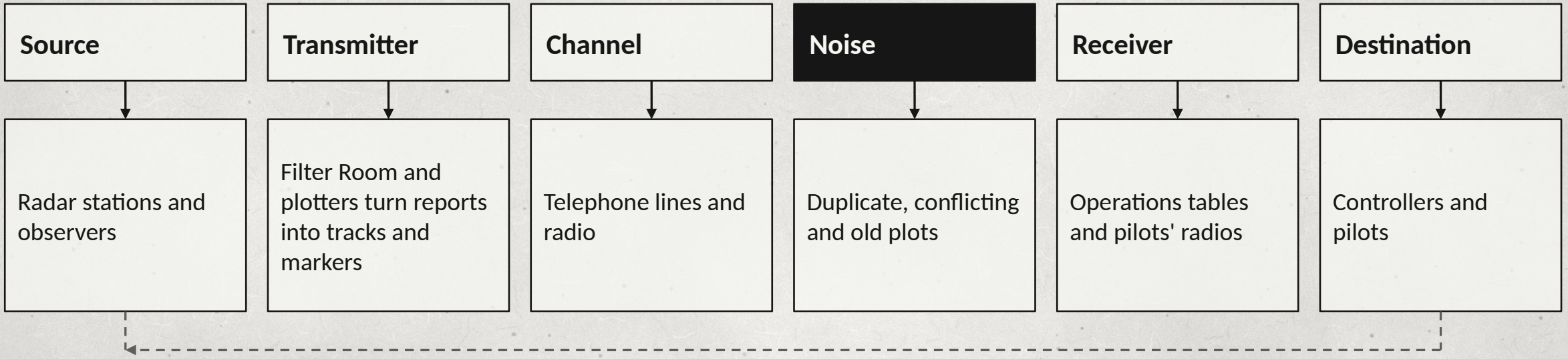
# The theories applied

*Does Dowding's network pass both theories, and does Amy's chat?*

3

# The network kept noise out of the chain

*Each part of Shannon and Weaver's model matches a part of the network*




*Feedback: radio position fixes and pilots' reports. The original model has none.*

**The difference from Amy's chat** The network removed noise before anyone read the table, and every level knew who acts. In Amy's chat, every unfiltered message reaches everyone, and nobody knows who acts.

Shannon & Weaver (1949); Dowding (1946); Imperial War Museums (n.d.-b)

# The radio order passes; Amy's request fails

*"Scramble. Vector 230. Angels 20." against "Can someone do the theory part?"*

| | Radio order | Amy's request | |
|---|---|---|---|
| **Clarity** | ●●● | ●○○ | *Amy names no one* |
| **Correctness** | ●●● | ●●● | |
| **Conciseness** | ●●● | ●●● | |
| **Concreteness** | ●●● | ●○○ | *No deadline or detail* |
| **Completeness** | ●●● | ●○○ | *Nothing to act on* |
| **Consideration** | ●●● | ●○○ | *The reader must guess* |
| **Courtesy** | ●○○ | ●●● | *Amy wins* |

## 19 vs 13

The order wins on everything except courtesy. Amy wins only on courtesy.

*The 7 Cs give every quality equal weight, but the situation decides which ones matter. In a cockpit, courtesy could wait. In a group chat, it cannot.*

Dowding (1946); Murphy et al. (1997). Scores are our group's judgment.

# Tomorrow's Leaders

*Part 4*

## Lessons for us

4

*Amy's team cannot get more members or more time.*
*How can it waste less?*

# Practice 1: One task board for the project

*Set it up like the operations table: one board, one owner per card, the same few columns, every deadline in view*

1. **One board for the whole project**
   Every task is a card. If it is not on the board, it does not exist.

*1940: one table for every level*

2. **One owner and one due date per card**
   "Theory slides, Ming, Thursday."

*1940: the raid block*

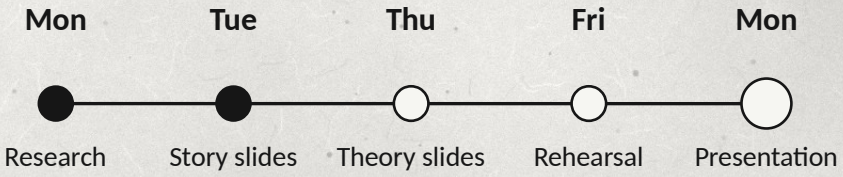
3. **Four columns**
   Not started, Drafting, Ready for review, Done.

*1940: the tote board*

4. **A roadmap of deadlines**
   Every deadline on one timeline, so the team sees the time left.

*1940: the 20-minute limit*

**Amy's roadmap**




*Filled: done. Hollow: still to come.*

**Evidence** In a flight simulation study, pairs who shared an understanding of the task and of each other worked together better and performed better.

**Where it works** Free tools such as Trello, Notion or Microsoft Planner, in any team that shares work.

Mathieu et al. (2000)

# Practice 2: Pass updates through the board

*Follow the network's path: report once, filter once, then send each request to one person*

1. **Report once**
   Update the card. Do not repeat it in the group chat.

   *1940: one network*

2. **Filter once**
   Twice a week, the coordinator merges duplicate cards and closes old ones.

   *1940: the Filter Room*

3. **Assign, do not ask**
   Mention one person on the card: "@Ming, theory slides by Thursday 8 pm, please."

   *1940: an order to one squadron*

4. **Confirm**
   The owner replies "Got it" or moves the card to Drafting.

   *1940: pilots reported back*

### In the group chat

> *Amy: Can someone do the theory part?*

**Five people read it. Nobody acts.**

### On the board

**Theory slides**
@Ming · Thu 8 pm · Drafting

**One person reads it. One person acts.**

**Evidence** Simpler letters to 35,050 tax filers led more of them to claim a benefit they were eligible for. Listeners who paraphrased made speakers feel more understood.

**In Hong Kong** Hall would call Hong Kong more high-context, though his model is rarely tested. Keep the "please" when you assign a card.

Bhargava & Manoli (2015); Weger et al. (2014); Hall (1976); Cardon (2008)

# Tomorrow's Leaders

## Germany had more aircraft. Britain wasted less.

*Your team cannot get more members or more time. It can waste less, when the right information reaches the right person in time.*

- One board, read by everyone
- Update the card, not the chat
- Assign each task to one person

### Questions for you

*Where does information get lost in your group projects?*

*Does your team have a board, or only a group chat?*

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